

Class #6 - Span and Geometry

1. Each vector $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \underline{\hspace{2cm}}$ can be visualized as an arrow starting from the $\underline{\hspace{2cm}}$ and terminates at the $\underline{\hspace{2cm}}$ which has the $\underline{\hspace{2cm}}$ (x_1, x_2) . There is a $\underline{\hspace{2cm}}$ correspondence between the two expressions $\underline{\hspace{2cm}}$ and $\underline{\hspace{2cm}}$, so we treat them the same.

Example: $u = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$. Sketch $-u, u + (1/2)u$.

What will be $\{cu : c \in R\}$? $\underline{\hspace{4cm}}$

Note: The set $\{cu : c \in R\} =$ The set of all $\underline{\hspace{4cm}}$ of u .

2. Def: Given $u_1, u_2, \dots, u_p$ in $\mathbf{R}^n$, $\text{Span}\{u_1, u_2, \dots, u_p\}$ is the set of $\underline{\hspace{2cm}}$ linear combinations of the vectors $u_1, u_2, \dots, u_p$, i.e.,

$$\text{Span}\{u_1, u_2, \dots, u_p\} = \{c_1u_1 + c_2u_2 + \dots + c_pu_p : c_i \in R\}.$$

Example: $u + (1/2)u \in \underline{\hspace{4cm}}$, and

$u + (1/2)v \in \underline{\hspace{4cm}}?$

3. Example: $u = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad v = \begin{bmatrix} -4 \\ 3 \end{bmatrix}.$

$u + v$ is the $\underline{\hspace{2cm}}$ of the $\underline{\hspace{2cm}}$ whose other vertices are $v, 0$, and $\underline{\hspace{2cm}}$.

$u - v = \underline{\hspace{2cm}}.$

What is $\text{Span}\{u, v\}$?

4. Example: $u = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad v = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$

$\text{Span}\{u, v\} =$ _____

$\text{Span}\{0\} =$ _____.

5. Def: We say that two vectors u, v in $\mathbf{R}^n$ are *dependent* if one of the vectors is a constant multiple of the other vector.

If two vectors are not dependent, we say u and v are _____.

6. Examples:

(a) 0 and u are _____ for every $u \in \mathbf{R}^n$.

(b) In $\mathbf{R}^3$,

$\text{Span}\{u\} =$ the _____ through u and 0 if u _____.

$\text{Span}\{u, v\} =$ the _____ containing _____ if u and v are _____.

$\text{Span}\{u, v, w\} =$ _____ if _____.

(c) Let $a_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 0 \\ 3 \\ 6 \end{bmatrix}, \quad a_3 = \begin{bmatrix} -4 \\ -2 \\ 3 \end{bmatrix}, \quad \text{and} \quad b = \begin{bmatrix} 4 \\ 1 \\ -4 \end{bmatrix}, \quad .$

Let $W = \text{Span}\{a_1, a_2, a_3\}$.

(a) Is $b \in \{a_1, a_2, a_3\}$? _____ How many vectors are in $\{a_1, a_2, a_3\}$? _____

(b) Is $b \in W$? _____

How many vectors are in W ? _____

(c) Is $a_1 \in W$?

(d) Are any two of the vectors a_1, a_2 and a_3 dependent?

(e) Describe the set W geometrically.